

Te-Hsun Tsai

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EDUCATION

University of Washington

Seattle, WA | Sep. 2025 – Expected Mar. 2027

Master of Science in Statistics (Advanced Methods & Data Analysis) (GPA: 3.88/4.0)

Ph.D.-level Pass — UW Statistics Theory Exam

National Chengchi University

Taipei, Taiwan | Sep. 2021 – Jun. 2025

Bachelor of Science in Statistics (GPA: 3.88/4.0)

RESEARCH INTERESTS

Reliable ML under Distribution Shift | Uncertainty Quantification & Conformal Prediction | Sequential & Multimodal Learning

SUBMITTED MANUSCRIPT

Chien, F. P.*, Li, V. C.-S.*, Tsai, T.-H.*, et al. *Co-first authors (equal contribution).

“Decoding Prior-Induced Speech States from fMRI under Identical Acoustics.” *Submitted to IEEE ICASSP 2027.*

RESEARCH EXPERIENCE

Research Intern — Machine Learning for Speech & Neural Data

Taipei, Taiwan

Academia Sinica | Advisor: Prof. Yu Tsao

Jun. 2026 – Sep. 2026

Prior-Induced Speech-State Decoding

- Developed a bidirectional S5 model with ROI-specific voxel projections for high-dimensional, small-sample fMRI decoding under acoustically identical speech stimuli.
- Built reproducible evaluation and model-comparison pipelines spanning SVM, residual MLP, Transformer, and S5 models with fold-specific preprocessing and multiple-comparison-corrected statistical testing.
- Evaluated cross-condition generalization to unseen Clean responses, achieving 64.29% held-out Pre/Post balanced accuracy and up to 68.45% Clean-to-Post projection.
- Linked Clean-to-Post projection to sentence-level comprehension gains (Spearman ρ up to 0.632, BH-FDR $q = 0.0014$), providing behavioral validation of the learned neural-state boundary.

Speech-Brain Representation Alignment

- Extracted hidden-state representations from pretrained speech encoders (WavLM, Wav2Vec2, Whisper) and language models (GPT-2, Phi-4), and quantified their alignment with EEG and fMRI responses across clean and noisy speech conditions.
- Evaluated layer-wise model-brain alignment using ridge regression, leave-run-out cross-validation, Pearson correlation, noise-ceiling normalization, and robust subject- and channel-level aggregation.
- Compared representation alignment across model families, Clean/Noisy conditions, neural modalities, and functional subsets to characterize robustness under heterogeneous data conditions.
- Identified representations with stable neural alignment across heterogeneous conditions, revealing which model layers generalized most consistently across datasets and modalities.

Research Assistant — Cause-of-Death Modeling

Seattle, WA

University of Washington | Research Supervisor: June Yang, Research Scientist

Sep. 2025 – Dec. 2025

- Developed machine-learning benchmarks using SVM and Random Forest for automated cause-of-death classification across heterogeneous verbal-autopsy datasets with highly imbalanced, long-tailed labels.
- Designed hierarchical classification architectures with soft pruning and chapter-to-detail constraints, and performed systematic hyperparameter tuning across age-stratified cohorts.
- Built strict no-leakage evaluation and multimodal feature-fusion pipelines for heterogeneous clinical data, emphasizing reliable evaluation under sparse, heterogeneous, and small-sample settings.

Undergraduate Researcher — Forecasting & Data Assimilation for Chaotic Systems

Taiwan

National Center for Theoretical Sciences (NCTS) | Advisor: Prof. Feng-Nan Hwang

Jul. 2024 – Aug. 2024

- Developed hybrid forecasting methods combining LSTM/ARIMA predictors with Kalman-filter data assimilation for nonlinear Lorenz-63 dynamics.
- Conducted 200+ controlled experiments evaluating forecast robustness across training-data size, forecast horizon, observation noise, and model misspecification.

SELECTED RESEARCH PROJECT

Adaptive Conformal Prediction for Chaotic Time Series — Graduate Research Project

- Studied split conformal prediction under closed-loop chaotic forecasting, demonstrating substantial coverage degradation as forecast horizon increased.
- Evaluated rolling-window and EWMA adaptive calibration strategies, recovering near-nominal long-horizon coverage while characterizing the coverage-interval-width trade-off.

PROFESSIONAL EXPERIENCE

Optimization & Scheduling System Engineer

Taipei, Taiwan

Elementary Schools

Jan. 2025 – Jun. 2025

- Developed and deployed an integer linear programming scheduling system optimizing 152 course-teacher assignments under policy and availability constraints, reducing administrative workload by 80%+ through automated validation and FET-compatible workflows.

TECHNICAL SKILLS

Programming & Tools: Python, R, SQL, MATLAB, C++, SAS, Stata, LaTeX, Git

Machine Learning: PyTorch, Transformers, S5/Mamba, SVM, Random Forest, XGBoost

Reliability & Statistical Learning: Conformal Prediction, Uncertainty Quantification, Robust Evaluation

Sequential & Multimodal Data: LSTM, ARIMA, Kalman Filtering, Data Assimilation, EEG/fMRI, Neural Encoding, Speech-Brain Modeling